Pilotwings 64 (US) save file (EEPROM)

As of 20 October 2026. From src/app/save.c, code_B2900.c, level_select.c and test_menu.c of the decompilation, checked against a real save file.

File

- 4 kbit EEPROM = 512 bytes (some emulators store 2048; the game uses only the first 512).
- Two saved games of 256 bytes: File 1 at 0, File 2 at 256.

One saved game (256 bytes)

Bytes / bits	Content
bytes 0-1	magic: "PW" = saved game, "pw" = empty (prepared by the game)
from bit 16	results: one 7-bit value per test, 57 tests in the retail game (61 with the game patch bonus-birdman-v1)
behind them	photo album: 6 photos × 172 bits
up to byte 254	unused (zeros)
byte 255	checksum: sum of bytes 0-254, low 8 bits (not set for empty files)

All values are a bit stream; bit n of a byte is 1 << n.

Order of the results: for every vehicle (Hang Glider, Rocket Belt, Gyrocopter, Cannonball, Sky Diving, Jumble Hopper, Birdman), for every class (4 for the license vehicles, 3 for the others), for every test. The number of tests comes from the mission files (UPWT, COMM block: class, vehicle, test), so hacks with other missions get another layout. Retail: license vehicles 1/2/3/3 tests per class; Cannonball 3 levels × 4 targets; Sky Diving and Jumble Hopper 3 levels × 1; Birdman 3 × 4. Birdman stage 4 is not saved.

With the game patch "Bonus games and Birdman" (bonus-birdman-v1): saveFileWrite and saveFileLoad save 3 classes only for Cannonball, Sky Diving and the Jumble Hopper and 4 for all others, so Birdman stage 4 is saved too (4 more results, 28 bits; 61 results). Everything up to Birdman stage 3 stays where it was, but the photo album starts 28 bits later. A save file of the game without the patch therefore shows wrong photos in a patched game (and the other way round); the results are read correctly. The editor's save file tab uses the layout of the open project.

Values: 0-100 points (Cannonball: 0-25 per target, 100 per level); 127 = no result yet. Sums count 127 as 0.

Photo record (172 bits): camera position (3 × s16), direction (3 × u8, × 1.40625°), test (2 bits), class (2), object types (4 bits each), animation time per object (u8), vehicle (3), object count (3),

object instance (3 bits each). Object types: 1 shuttle, 2 ferry, 3 missile, 4 whale, 5 fountain, 6 flame, 7 whale pod, 8 plane, 9 glider, 10 boat, 11 nothing. Without the fix photo-album-v1 the instance numbers of objects 5 and 6 are lost.

Rules (what is open)

Medal limits (bronze/silver/gold): Beginner 70/80/90, Class A 140/160/180, Class B and Pilot 210/240/270, bonus games 70/80/90 per level. The Birdman's entry (index 7, medal index = vehicle + 1) is 0/0/0 in the game; the game patch bonus-birdman-v1 sets it to 280/320/360 per stage. A license class has a medal only if none of its tests is 127.

- **Class A:** all three license vehicles have a medal in Beginner.
- **Class B / Pilot:** per vehicle, a medal in the class before.
- **Cannonball / Sky Diving / Jumble Hopper:** all three license vehicles have silver in Class A / B / Pilot. Level 2 and 3: at least 70 points in the level before.
- **Birdman stage 1:** silver in Beginner with all three vehicles. Stages 2-4: silver in Class A/B/Pilot with all three, and at least 80 points in every level of Cannonball / Sky Diving / Jumble Hopper.

Birdman results open nothing and do not count for "all perfect" (also with the patch; to be decided when the Birdman gets points).

The editor's save file tab (Hawk's Hangar) follows these rules. With the patch it shows the points of the 16 Birdman missions (4 stages × 4) below the Birdman row; a stage's fields can be edited when the stage is open.